

DRY-RUN GUIDE | ss233.py

Source: Python\3rd\_Belt\ss233.py

#### CORE BEHAVIOR

Run-length encodes consecutive character groups, but returns the original unless encoding is strictly shorter.

#### REPRESENTATIVE INPUT

'aaabbc'

#### STEP TRACE

Runs become a3,b2,c1 -> 'a3b2c1', length 6; input length is also 6, so retain original.

#### EXPECTED OUTPUT / RETURN

'aaabbc'

#### ACTUAL SOURCE EXCERPT

Verbatim excerpt from Python\3rd\_Belt\ss233.py; line numbers refer to the original file.  
No source was modified.

```
001 | # Question: Solve the given array/string challenge using the required logic.
002 | # Add your solution here.
003 |
004 | def compress_string(s):
005 |     # Handle empty string edge case
006 |     if not s:
007 |         return s
008 |
009 |     compressed = []
010 |     current_char = s[0]
011 |     count = 1
012 |
013 |     # Iterate through the string starting from the second character
014 |     for i in range(1, len(s)):
015 |         if s[i] == current_char:
016 |             count += 1
017 |         else:
018 |             # Append the character and its consecutive count
019 |             compressed.append(current_char + str(count))
020 |             current_char = s[i]
021 |             count = 1
022 |
023 |     # Append the last character group
024 |     compressed.append(current_char + str(count))
025 |
026 |     # Join the list into a single compressed string
027 |     compressed_str = "".join(compressed)
028 |
029 |     # Return compressed string only if it is strictly shorter than the original
030 |     if len(compressed_str) < len(s):
031 |         return compressed_str
032 |     else:
033 |         return s
034 | input_string = input()
035 | result = compress_string(input_string)
036 | print(result)
```

#### DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.